SANSKAR SHARMA

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PROFESSIONAL SUMMARY

Final-year Mechatronics Engineering student with extensive hands-on experience in embedded systems, FreeRTOS concurrent design, IoT architectures, and sensory integration. Experience in developing fault-tolerant hardware platforms, real-time cloud synchronization (MQTT/Firebase), and adaptive multi-sensor monitoring systems. Driven to engineer robust, enterprise-grade embedded IoT hardware solutions.

TECHNICAL SKILLS

- **RTOS & Firmware:** FreeRTOS (Task Scheduling, Queues, Semaphores, Mutexes, Watchdog Timers), Embedded C, Arduino IDE, N8N
- **Microcontrollers & Hardware:** ESP32, Arduino UNO, MSP430, Servo Motors, HX711 Load Cell, MQ2, MPU6050, OLED, GSM
- **Protocols & Cloud:** MQTT, Firebase, SPIFFS, I2C, SPI, UART, GSM/GPRS
- **Tools & Simulation:** SolidWorks, AutoCAD, Proteus, HTML/CSS
- **Languages:** English (Proficient), Hindi (Native), French (Beginner)

WORK & LEADERSHIP EXPERIENCE

IoT Engineering Intern — *Elegance Skills (Remote)* **Jul 2026 – Aug 2026**

- **FreeRTOS Concurrent Architecture:** Designed a real-time hospital monitoring system using FreeRTOS on ESP32, managing multi-sensor readings, OLED displays, and MQTT publishing via priority-based task scheduling, mutexes, and queues.
- **Adaptive Sampling & Power Optimization:** Developed dynamic sampling logic (vTaskDelay) that automatically shifts reading frequency from 60s to 5s during critical vital spikes to preserve battery life while ensuring safety.
- **Fault-Tolerant Offline Buffering:** Built a network fault detection system using SPIFFS/queue buffering with exponential backoff reconnects to eliminate data loss during Wi-Fi or MQTT outages.
- **Control Systems & Dual Dashboards:** Implemented smooth servo-driven bed elevation (0–90°) and safety-restricted dosage control (0–100 mg/hr) with dual role-based MQTT dashboards for medical and facility staff.

Embedded System Intern — *Reemzet Solutions LLP (Gaya, Bihar)* **Jun 2026 – Jul 2026**

- **IoT Smart Shopping Cart:** Engineered an ESP32-based contactless billing cart integrated with Firebase/MQTT for real-time cloud inventory updates and mobile app checkout.
- **Anti-Theft System:** Designed a real-time weight verification safety mechanism using an HX711 load cell and buzzer to flag unscanned items.

Google Student Ambassador -- Remote **Jul 2025 – Jan 2026**

- Conducted hands-on technical workshops on 'Automation using Agentic AI' (N8N) and represented Google tech initiatives across university cohorts.

KEY ENGINEERING PROJECTS

Smart IoT Safety Platform for Industrial Workers — *Embedded Systems / IoT* **Feb 2026 – Present**

- Built a low-cost wearable device using ESP32, MPU6050 (6-axis gyro), and MQ2 gas sensors to monitor environmental hazards, gas leaks, and worker fall detection in real-time.

Electric Fence Unauthorized Tampering System — *Smart India Hackathon 2025* **Sep 2025 – Oct 2025**

- Solved a Kerala State Govt challenge by designing an IoT security system using Arduino UNO, LM393 voltage, and ACS712 current sensors with instant GSM alerts upon wire tampering.

EDUCATION

B.Tech in Mechatronics Engineering *(CGPA: 8.0 / 10)* **2023 – 2027**
Puducherry Technological University

Class 12th Board (CBSE) - 76% | Class 10th Board (CBSE) - 86% **2019 – 2022**
Trident Public School | Sunshine Prep High School

HONORS & CERTIFICATIONS

- **Naukri Young Turks (Mechanical Engineering):** Secured All India Rank (AIR) 1605.
- **Certifications & Job Simulations:** NPTEL Forests & Management (IIT Kanpur), AWS Solutions Architecture Job Simulation, Citi Wealth, Google Arcade Novice.